

AI in Retail and Supply Chain

Plan inventory. Personalize discovery. Prepare accurate product listings.

Predict a number

How many units of this product will sell next week, so a planner can order early enough.

> Put things in order

Which ten of 4,443 products deserve this shopper's screen.

> Read and draft

Turn a supplier document into catalog fields, and hold whatever the evidence cannot support.

A retailer makes the same five decisions about every product

Source

Choose which products to sell and which suppliers to buy them from.

> Stock

Decide how many to buy, where to hold them, and when to reorder.

> Sell

Price the product, put it in front of the right shopper, and describe it accurately.

> Fulfill

Pick it, ship it, and get it to the customer when promised.

> Support

Handle returns, complaints and the feedback that follows.

THREE GOALS THAT PULL AGAINST EACH OTHER

Availability

Every stockout is a sale you never see, so the safe move is to hold more. But stock is cash sitting on a shelf, and in this business it can sit there for a season. Watched as: stockout rate and inventory days.

Margin

A discount reliably moves more units. It just as reliably moves less profit per unit, and a promotion that sells out the shelf can still lose money. Watched as: gross margin and markdown waste.

Trust

The thing in the box has to be the thing on the page. When it is not, you pay twice: the return, and the customer who stops believing the next listing. Watched as: returns, complaints and corrections.

Seven places AI shows up, and the three we build tonight

ACTIVITY	THE QUESTION SOMEBODY IS ACTUALLY ASKING	HOW IT IS USUALLY DONE	TONIGHT
Assortment	Which products should this store carry at all next season?	Demand analysis plus optimization against shelf and budget limits	
Inventory	How many units of this product will sell next week?	Regression on the product's own sales history	CASE 1
Pricing	What discount moves units without giving away the margin?	Price-response models and controlled experiments	
Discovery	Which products belong in this shopper's ten slots?	Recommendation from what similar shoppers bought	CASE 2
Catalog operations	Which product facts are supported well enough to publish?	LLM extraction from supplier documents, then rule checks	CASE 3
Store execution	Which shelves are empty right now?	Computer vision on shelf images plus stock signals	
Fulfillment and returns	Where should this order ship from, and why did it come back?	Routing optimization, anomaly detection, text analysis	

Start from the decision, not the technology. Three of these seven rows are not machine learning at all, and one of the three we build is not a model we train.

Tonight's route, and what the big retailers have published

THE PLAN

Retail context

Value chain, the AI map, and what the data hides

Case 1 / Forecasting

One real product, four methods, 26 held-out weeks

Case 2 / Next Best Product

Shared buyers, saved links, one customer's real list

Case 3 / Catalog

Source facts, a captured draft, checks and holds

Compare and apply

Three outputs, one set of design questions

120 minutes of planned content plus a break. No live service, sign-in or network call is required tonight.

REAL SYSTEMS, DESCRIBED BY THE COMPANIES THEMSELVES

Amazon, item-to-item filtering

Their engineers published the method that ranks products from shopping behavior. Case 2 is a small version of it.

Walmart, demand and pricing

Published research on forecasting and price experiments. Case 1 produces the same kind of numeric input to planning.

Amazon seller listings, 2023

Announced that LLMs draft titles and descriptions which sellers then review. That review step is case 3.

Walmart Trend-to-Product, 2025

Described generative AI proposing product concepts that designers and merchants then refine. People stay in the loop.

These establish that the patterns are real. They do not transfer anyone's results to our classroom numbers.

Case 1: how many of these bags will sell next week?

Product photo
not included
in public copy

Stock code 85099B · week beginning 21 November 2011



The size of the miss

Average: off by 634 units.

Model: off by 71 units.

The eight weeks behind this one included the Christmas ramp, so the average was dragged up. Averages lag a turning point.

JUMBO BAG RED RETROSPOT

The planner has to guess first

The order goes to the supplier before anyone knows what the week will do. A forecast is an estimate made deliberately early, which is the only time it is useful.

Units, never money

The model predicts units of one product in one week. It says nothing about revenue, margin or whether stocking this bag was a good idea at all.

A forecast is not an order

Nothing here places a purchase order. The number is one input to a person who also knows lead times, case packs and the promotion calendar.

From 197,951 invoice lines to one row per product-week

Invoice lines

One row per product per order, exactly as the shop recorded it. 197,951 rows.

> Clean sales

Drop cancellations, returns and non-product lines such as postage and fees.

> Weekly totals

Add up units by product and by week. 4,871 products across 102 weeks.

> Modeling panel

Keep the 469 products that sold steadily enough to have a learnable history.

TIME RUNS FORWARD AND IS NEVER SHUFFLED

8 weeks

Seed the history

Only here to build the history the first real row needs.

52 weeks

Fit the model

The model learns its rules here. 469 products x 52 weeks = 24,388 training rows.

16 weeks

Choose the method

Four candidates are compared here, and only here. The winner is frozen.

26 weeks

Final check, opened once

Never seen during any choice. 12,194 rows. This is the only honest score.

Dec 2009

Nov 2011

UCI Online Retail II | DOI 10.24432/C5CG6D | CC BY 4.0 | one UK gift wholesaler, December 2009 to December 2011. Recorded sales, not unmet demand.

A row may only contain what was knowable before that week

WHAT THE MODEL GETS	THE ACTUAL COLUMNS	WHAT IT LETS THE MODEL NOTICE
Recent history	lag1, lag2, lag3, lag4, lag8	Units sold one, two, three, four and eight weeks ago. Raw memory of what this product has been doing.
Recent level	ma4, ma8	The four-week and eight-week averages. Where this product normally sits when nothing unusual is happening.
Recent volatility	std4	How much the last four weeks bounced around. Some products are steady; some are not, and the model should treat them differently.
Position in the year	lag52, week number, sine and cosine of the week	What happened this same week last year, and where we are in the annual cycle. December is not July.

ONE REAL ROW · RED RETROSPOT · week of 6 June 2011

lag1 = 273 units ma4 = 847.5 ma8 = 665 → the model must guess → the week actually sold 883

The target never appears in its own row, and nothing from a later week does. Product identity is excluded too, so one model serves all 469 products.

One model for all 469 products, chosen by trying four

469 products, one model

One product has only 52 training weeks, far too few to learn from. Pooling every product gives 24,388 rows and one shared set of rules.

Product identity is not an input

The model never learns that row 12 is the red bag. It learns what a rising four-week average tends to mean, which is a pattern that transfers to products it has never seen.

The choice is made once

Four methods compete on the same sixteen validation weeks. The winner is refit on all 31,892 pre-test rows and then frozen before the test weeks are opened.

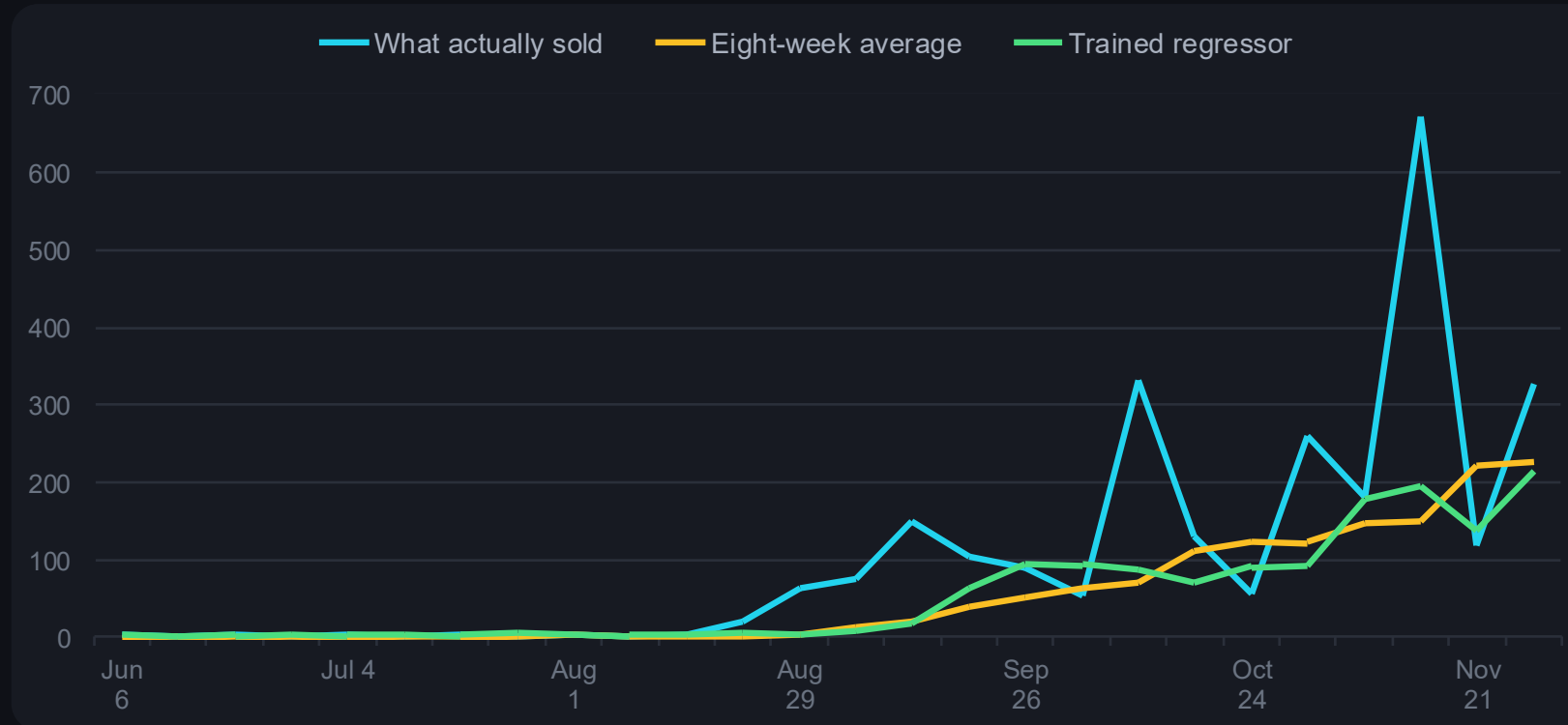
VALIDATION MAE · average miss in units, on sixteen weeks no candidate was fitted on · lower is better



Absolute-error loss aims at the typical week instead of chasing the rare huge week. Slide 11 shows what that choice costs.

The forecast follows the trend and still lags a real surge

RED WOOLLY HOTTIE WHITE HEART, a knitted hot water bottle cover, across all 26 held-out weeks.



Summer: easy

One to four units a week. Every method gets this right, because predicting "about two" needs no skill at all.

November: not easy

The peak week sold 673. The model said 195, the average said 151. Both saw the season coming. Neither sized it.

The model reacts faster than the average because it can use last year and the shape of the rise. It still cannot size a peak it has only seen twice in the whole dataset.

A smaller average miss, and a planner who still decides

FROZEN TEST RESULT · 12,194 held-out product-weeks, both methods scored on exactly the same rows

MAE · the average size of a miss

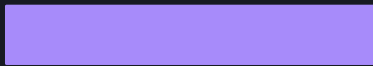
Eight-week average  **52.31**

Regressor  **47.97**

Better by 8.3%. This is the typical week.

RMSE · the same misses, with large ones punished

Eight-week average  **129.75**

Regressor  **133.78**

Slightly worse. The big misses did not improve.

AND THEN A PERSON TURNS IT INTO AN ORDER

825

Forecast, rounded
up

+100

Safety stock

-300

Already on the
shelf

-200

Already on its way

425

What to actually
order

Still the planner's call

Case packs, lead times and next month's promotion are all outside the model. It contributes one number.

[Open the forecasting notebook](#)

[Open the forecasting simulator](#)

Safety stock, on-hand and incoming are classroom assumptions.

Case 2: ten slots, 4,443 products, one shopper

Popularity

Count how many customers bought each product during training and show the winners to everyone. No personalization at all. It is a genuinely strong baseline and it never fails to produce ten products.

Item-item recommendation

Start from what this person already bought. Find the products that other customers tended to buy alongside those. Add up the connections and rank what comes out on top.

The merchandiser's filter

Before anything is shown: drop what this customer already owns, drop anything blocked for stock or policy reasons, and rank only what survives. A person owns this list, not the model.

CALL IT NEXT BEST PRODUCT, NOT NEXT BEST OFFER

An offer implies we know this shopper will respond. This dataset has no record of what was ever shown to anyone, no record of what was ignored, and no version of history where a different product appeared instead. A ranking is a decision about where ten scarce slots go. It is not a claim about need, intent or probability of purchase.

[Open the recommendation visual walkthrough](#)

4,962 customers and 4,443 products in the training period, which ends 9 September 2011.

Two bags, 413 shared customers, one number

THE GRID THE MODEL LEARNS FROM

	Red bag	Pink bag	Lunch bag	Cake stand
Asha	1	1	1	0
Ben	1	1	0	0
Cara	1	0	1	1
Diego	0	0	0	1

Illustrative grid. The real one is 4,962 customers by 4,443 products and 98.2% empty.
A zero means "never saw it", not "did not like it".

Product photo not included in public copy

844 buyers

Rex London

Product photo not included in public copy

541 buyers

Rex London

413 bought both

$$413 / \sqrt{(844 \times 541)} = 0.611$$

The square root stops a bestseller looking similar to everything just because it sells.

What makes them similar

Nothing about the products themselves. Only this: the same people bought both. Overlap in buyers is the whole signal.

It cannot see these photos

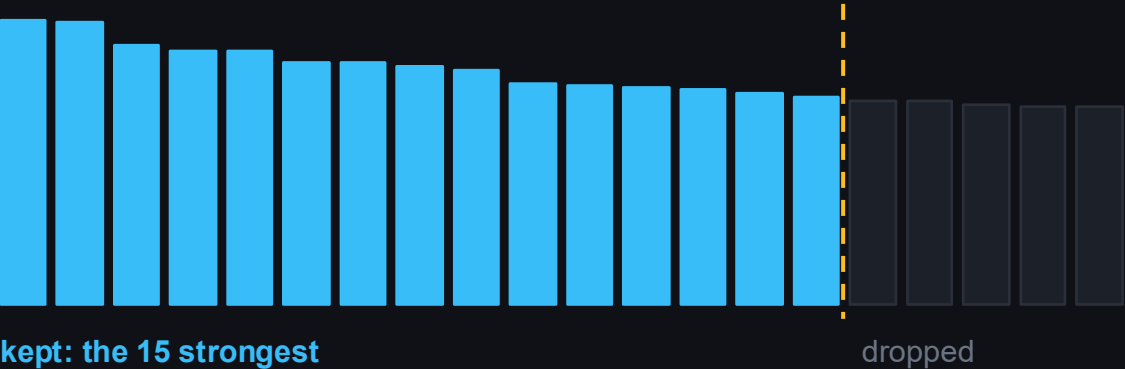
It does not know both are bags, both patterned, both the same shape. It knows 413 people bought both. That the two look alike is a coincidence that happens to confirm the method.

Training period ends 9 September 2011. Repeated purchases of the same product still count once.

Keep the fifteen strongest links per product, then add them up

TRAINING: COMPARE, DROP THE SELF-LINK, KEEP FIFTEEN

Links from JUMBO BAG RED RETROSPOT, strongest first



0.611 at the top, 0.438 by rank sixteen, and no real gap below that. Those weak links are mostly noise, and keeping them makes every list blander.

Is fifteen actually better than keeping every link? Slide 17 measures both.

PREDICTION: LAY DOWN THE LINKS AND ADD

A customer who bought the red bag and the strawberry bag

CANDIDATE	FROM RED	FROM STRAW.	TOTAL
Lunch bag	0.816	0.500	1.316 first
Cake stand	0.333	—	0.333 second
Red bag	already bought		excluded

Adding, not averaging

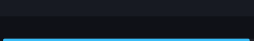
A total above 1 is normal and is not a probability. A product connected to several things you bought outranks one connected to a single thing, which is exactly the behavior we want.

Fictional teaching example: 6 products, 2 slots. The real model stores 15 links per product and fills 10 slots.

What customer 12349 actually sees

The trained model's real ten slots for one real customer, in rank order, with the actual scores.

RANK · PRODUCT · SCORE

1	PACK OF 72 RETROSPOT CAKE CASES		4.883
2	DOORMAT NEW ENGLAND		4.478
3	DOORMAT UNION FLAG		4.252
4	DOORMAT FAIRY CAKE		3.957 bought it later
5	DOORMAT FANCY FONT HOME SWEET HOME		3.832
6	WHITE HANGING HEART T-LIGHT HOLDER		3.732
7	RED RETROSPOT STORAGE JAR		3.454
8	PLASTERS IN TIN SPACEBOY		3.452
9	RED RETROSPOT TRADITIONAL TEAPOT		3.345
10	RED RETROSPOT BUTTER DISH		3.279

Product photo
not included
in public copy

Product photo
not included
in public copy

slot 2

slot 3

One hit in ten

Slot four was bought after the training cutoff. That is what the hit rate counts.

The nine are not wrong

One hit in ten slots is the normal shape of a working recommender.

A hit means the list contained something the customer later bought. It does not mean the list caused the purchase; this customer was already buying doormats.

A better hit rate, a wider catalog, and an honest fallback

THE THREE METHODS BEING COMPARED

One list for everyone

No personalization. Show the same training bestsellers to every shopper. The baseline to beat.

Personalized, all links kept

The item-item method from the last two slides, keeping every link a product has, strong and weak.

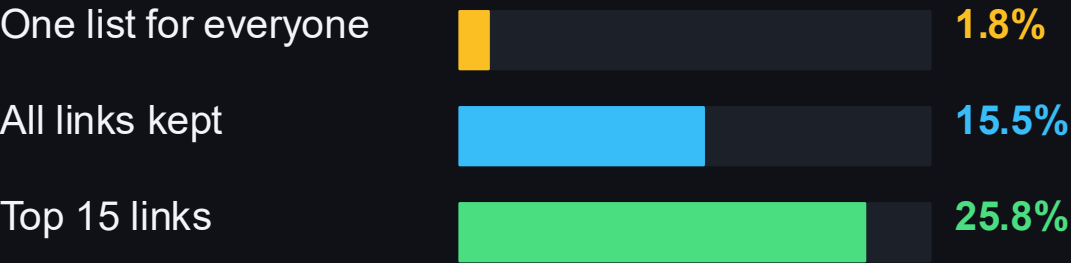
Personalized, top 15 links

The same method, but each product keeps only its fifteen strongest links. This is what we build.

HIT RATE AT 10 · share of 2,168 customers whose list held a later purchase



CATALOG COVERAGE · share of 4,443 products that reach anyone's ten slots



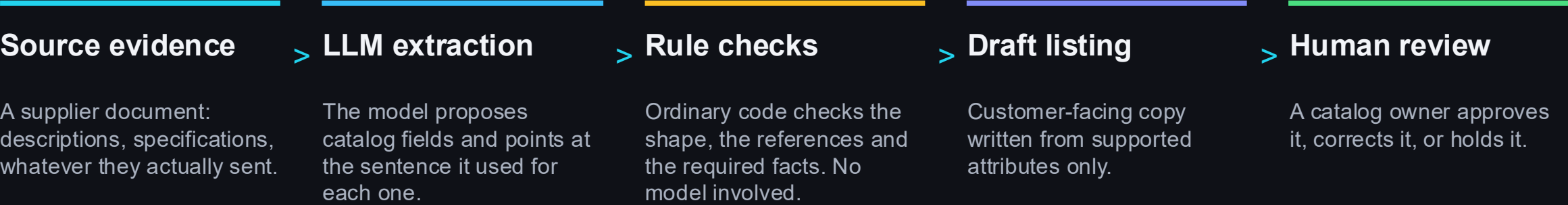
Throwing the weak links away beat keeping every one of them, on both metrics. Coverage matters too: at 1.8%, eighty products take every slot and 4,360 stay invisible.

[Open the recommendation notebook](#)

[Open the recommendation simulator](#)

Offline replay. It informed the model choice, and measures no revenue or causal lift.

Case 3: can this evidence become a published listing?



THREE WAYS THIS GOES WRONG, AND WHAT THE CUSTOMER EXPERIENCES

THE READING ERROR	WHAT ARRIVES AT THE CUSTOMER	THE QUESTION A REVIEWER ASKS
A shipping carton of six is read as a six-pack	They ordered what they believed was six items and one turns up	How many units does the customer actually receive?
A care or safety claim nobody wrote down	A ruined product, a return, or something genuinely unsafe	Which sentence in the source supports this wording?
Two sources give different dimensions	It does not fit the space they bought it for	Which specification is authoritative, and who says so?

All three are ordinary reading mistakes. A language model makes them fluently, in complete sentences, which is what makes them dangerous.

Five records: three real, two built to break it

3 real products

Dated public specifications with source URLs. Every proposed field traces back to a sentence somebody actually published, which is what makes checking possible.

Fictional case: missing fact

Six units per shipping carton, and the selling pack is simply never stated. The right answer is "unknown", and unknown is not the same as six or one.

Fictional case: contradiction

Two sheets for one item: 500 ml on one, 750 ml on the other. The right answer is to keep both and flag the conflict, not to quietly pick one.



PARTY BUNTING · 47566

Photograph: Rex London

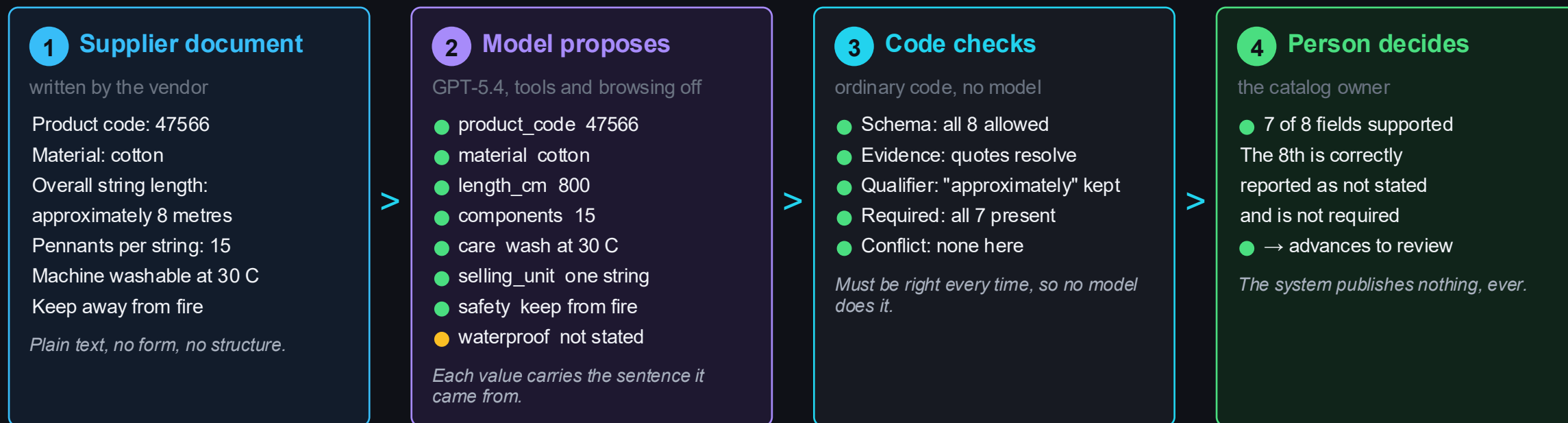
ONE REAL PRODUCT, TRACED FIELD BY FIELD

SOURCE SENTENCE	PROPOSED FIELD	THE QUALIFIER THAT MUST SURVIVE
"Cotton"	Material: cotton	Not a blend. The source says one fibre and only one.
"Approximately 8 metres"	Length: approx. 800 cm	The word approximately must survive the unit conversion.
"15 pennants"	Pennant count: 15	Fifteen flags on one string, not fifteen items in the box.
"Machine washable at 30°"	Care: machine wash at 30 C	Permission at one temperature, not a general claim.
"Keep away from fire"	Safety warning, kept as written	A warning. Not evidence of flame resistance.

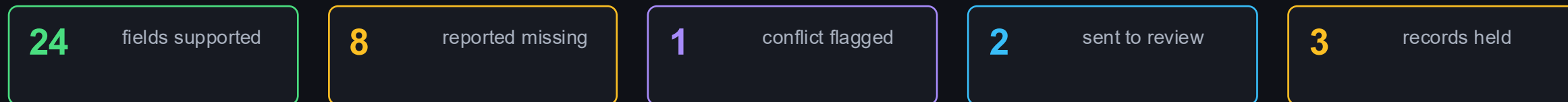
Official product evidence: Rex London. No prices, live stock, customer data or invented certifications enter the demonstration.

How one vendor document becomes a checked catalog record

Extract fields from the supplier's text, check every field against its source, then a person approves or holds.



ALL FIVE RECORDS TOGETHER



A hold is the system working, not failing. Five selected examples are a demonstration, not an accuracy benchmark.

A good draft, and three things it must not say

ACTUAL CAPTURED MODEL OUTPUT

Cotton bunting string

One string of cotton bunting with 15 pennants.
Overall string length is approximately 800 cm.
Machine washable at 30 degrees Celsius.
Keep away from fire.

GPT-5.4, captured 22 September 2026. Replayed offline; not edited for this slide.

WRITTEN BY US, TO SHOW WHAT MUST BE REJECTED

"Weatherproof"

No source mentions outdoor use at all.

"Flame-retardant"

The source says keep away from fire. This is the opposite claim.

"Guaranteed colorfast"

Nothing in the evidence says anything about colour holding.

THE ASYMMETRY THAT MAKES THIS HARD

Checking the left column means reading five source sentences. Checking the right column means noticing what is absent from them, which is far harder and is the first thing a reviewer stops doing at volume. That is why the checks are machinery rather than vigilance.

[Open the catalog-onboarding workflow \(offline\)](#)

Three different outputs, one set of design questions

	1 / FORECASTING	2 / NEXT BEST PRODUCT	3 / CATALOG ONBOARDING
The question	How many units of this product next week?	Which ten products go in this shopper's slots?	Is this evidence strong enough to publish?
What comes out	A number: predicted units	An ordering: ten products, ranked	Proposed fields, each pointing at its source
How it is judged	MAE 47.97 and RMSE 133.78 on 26 unseen weeks	Hit rate 41.6% and coverage 25.8% on later purchases	24 supported, 8 missing, 1 conflict across 33 fields
When it should refuse	A brand new product with no sales history to learn from	A customer with too little history: show "Popular right now"	A required fact nobody stated: hold the record and ask
Who is accountable	The inventory planner, who places the order	The merchandiser, who owns placement and exclusions	The catalog owner, who approves publication

None of tonight's results proves causal revenue lift, and none authorizes an unattended business action. Every measurement was offline replay on data the retailer had already collected.